TEXORA · LECTURE SHEET

Production Cost Statement and Break-Even Analysis

The full cost sheet built line by line from raw materials consumed to profit, and cost–volume–profit analysis: contribution, the P/V ratio, break-even in units and in sales value, margin of safety, and the break-even chart drawn exactly to the worked numbers.

AC-4205 · COST AND MANAGEMENT ACCOUNTING · TOPIC 6 · LEVEL 4 TERM 2

1. The statement of production cost (cost sheet)

Where Topic 4's job sheet costed one order, the **cost sheet** presents the cost of a whole period's production, stage by stage, each stage feeding the next. Its architecture is fixed: raw material consumed → prime cost → works (factory) cost → cost of production → cost of goods sold → total cost → profit against sales.

WORKED EXAMPLE — COST SHEET FOR A MONTH (20,000 SHIRTS SOLD)

Opening stock of raw material Tk 2,00,000 + Purchases Tk 18,00,000 − Closing stock Tk 2,50,000 = **Raw material consumed Tk 17,50,000**

Add direct labour Tk 5,80,000 and direct expenses Tk 70,000 → **Prime cost Tk 24,00,000**

Add factory overhead Tk 4,80,000; add opening WIP Tk 1,20,000, less closing WIP Tk 1,00,000 → **Works cost Tk 29,00,000**

Add administration overhead Tk 2,90,000 → **Cost of production Tk 31,90,000**

Add opening finished goods Tk 3,10,000, less closing finished goods Tk 2,60,000 → **Cost of goods sold Tk 32,40,000**

Add selling & distribution overhead Tk 1,60,000 → **Total cost Tk 34,00,000**

Sales: 20,000 shirts × Tk 200 = Tk 40,00,000 → **Profit Tk 6,00,000**.

Unit cost = 34,00,000 ÷ 20,000 = **Tk 170 per shirt**; profit per shirt Tk 30.

Read the sheet downward for totals and across for control: material is the dominant block, so a 1 % saving there (Tk 17,500) is worth more than a 10 % cut in direct expenses (Tk 7,000).

2. Break-even (cost–volume–profit) analysis

Break-even analysis applies the behaviour split of Topic 2. The engine concept is **contribution**:

Contribution per unit = Selling price − Variable cost per unit; P/V ratio = Contribution ÷ Sales

Break-even units = Fixed cost ÷ Contribution per unit; Break-even sales = Fixed cost ÷ P/V ratio

Margin of safety = Actual sales − Break-even sales; Profit = Margin of safety (units) × Contribution per unit

WORKED EXAMPLE — CVP FOR A SHIRT LINE

Selling price Tk 450. Variable cost per shirt: fabric Tk 210 + trims Tk 25 + making (CM) Tk 45 + packing Tk 20 = Tk 300.

Contribution = 450 − 300 = **Tk 150 per shirt**; P/V ratio = 150 ÷ 450 = **33.33 %**.

Fixed costs Tk 18,00,000 per period.

Break-even units = 18,00,000 ÷ 150 = **12,000 shirts**; break-even sales = 12,000 × 450 = **Tk 54,00,000** (check: 18,00,000 ÷ 33.33 % = Tk 54,00,000 ✓).

At actual sales of 16,000 shirts: margin of safety = 16,000 − 12,000 = **4,000 shirts = 25 % of sales**; profit = 4,000 × 150 = **Tk 6,00,000**.

For a target profit of Tk 9,00,000: required sales = (18,00,000 + 9,00,000) ÷ 150 = **18,000 shirts**.

3. The break-even chart, drawn to the same numbers

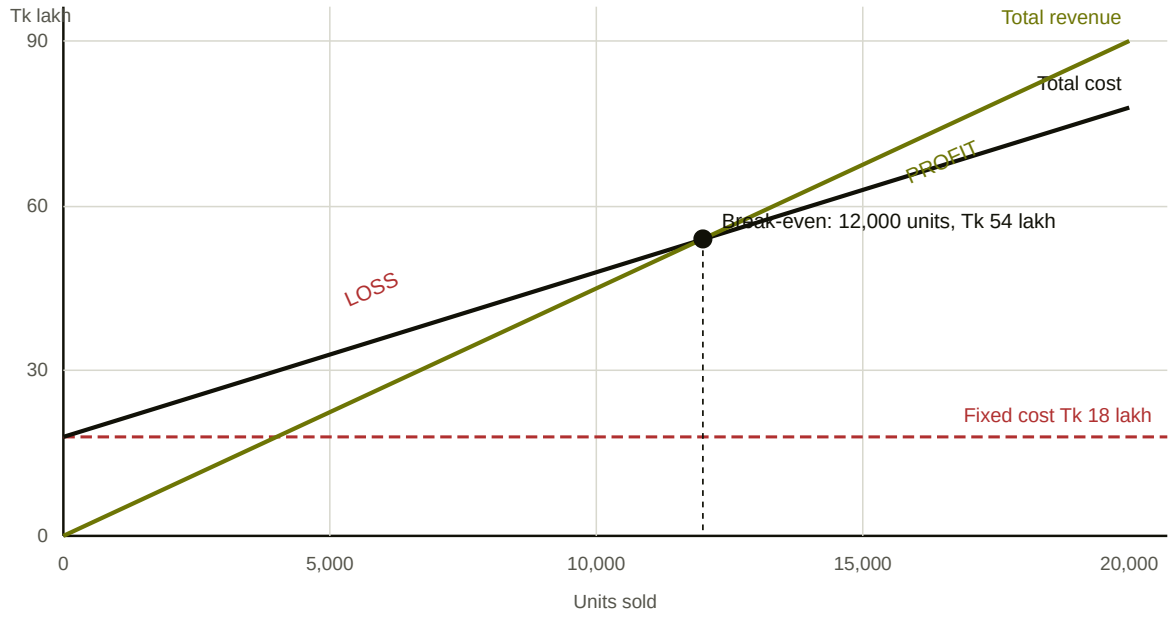


Figure 1 — Break-even chart for the worked example (price Tk 450, variable cost Tk 300, fixed cost Tk 18,00,000). The revenue and total-cost lines cross at 12,000 units / Tk 54,00,000; below the crossing the factory loses, above it each extra shirt adds its Tk 150 contribution to profit.

Assumptions and limits of the analysis: price and variable cost per unit constant, fixed cost constant within the relevant range, a single product or a constant sales mix, and production equal to sales. Multi-product firms apply the technique to a weighted-average contribution for the mix. Even where the assumptions bend, the framework disciplines thinking: every price cut, wage rise or machine purchase re-draws the chart, and management sees by how much.

Glossary

Cost sheet — the staged statement of a period's production cost. **Contribution** — price minus variable cost; what each unit contributes to fixed cost and profit. **P/V ratio** — contribution as a fraction of sales. **Break-even point** — sales level at which total contribution exactly covers fixed cost (profit zero). **Margin of safety** — the excess of actual over break-even sales; the fall sales can take before loss begins. **CVP analysis** — cost–volume–profit analysis.

Key points

• The cost sheet descends in fixed stages: consumed 17,50,000 → prime 24,00,000 → works 29,00,000 → production 31,90,000 → COGS 32,40,000 → total 34,00,000 → profit Tk 6,00,000 on sales of Tk 40,00,000; unit cost Tk 170.

• Contribution Tk 150 and P/V 33.33 % with fixed cost Tk 18,00,000 give break-even at 12,000 shirts / Tk 54,00,000.

• Margin of safety at 16,000 shirts is 4,000 shirts (25 %); profit = MOS units × contribution = Tk 6,00,000; target profit Tk 9,00,000 needs 18,000 shirts.

Practice

1. A product sells at Tk 600 with variable cost Tk 420 and fixed costs of Tk 9,00,000. Find contribution, P/V ratio, break-even units and break-even sales value, and the profit at 7,000 units.
2. From the cost-sheet example, recompute the cost of production if closing finished-goods stock were Tk 3,60,000 instead of Tk 2,60,000 (all else unchanged), and state what happens to total cost and profit if selling overheads are unchanged.
3. A firm sells at P/V ratio 40 % with fixed costs Tk 12,00,000. Find break-even sales value, and the sales needed for a profit of Tk 6,00,000.
4. List the four CVP assumptions and one practical situation violating each.

Answer key

1. Contribution = 600 − 420 = **Tk 180**; P/V = 180 ÷ 600 = **30 %**; BEP units = 9,00,000 ÷ 180 = **5,000 units**; BEP sales = 5,000 × 600 = **Tk 30,00,000** (check: 9,00,000 ÷ 0.30 ✓). Profit at 7,000 = (7,000 − 5,000) × 180 = **Tk 3,60,000**.
2. Cost of production is unaffected by finished-goods stock (it is computed before that line): still **Tk 31,90,000**. Cost of goods sold = 31,90,000 + 3,10,000 − 3,60,000 = **Tk 31,40,000** — Tk 1,00,000 lower than before, because a larger closing stock carries more cost forward. Total cost falls to Tk 33,00,000 and profit rises by Tk 1,00,000 to **Tk 7,00,000**. (If the question had meant closing WIP, works cost and everything below it would move instead — read the stage names carefully.)
3. BEP sales = 12,00,000 ÷ 0.40 = **Tk 30,00,000**. Sales for Tk 6,00,000 profit = (12,00,000 + 6,00,000) ÷ 0.40 = **Tk 45,00,000**.
4. Constant price — violated by quantity discounts and clearance sales. Constant variable cost per unit — violated by overtime premiums at high volume or bulk-buying discounts. Constant fixed cost in the range — violated when output forces a second shift (step cost). Constant mix / single product — violated when the order book shifts between high- and low-margin styles. (Any sensible pairing accepted.)